# Review for Final Exam

CMPT 365

Instructor: David Chou

- The final exam is cumulative
  - cover everything from the beginning of the class

NO HISTORY.

- This review mainly focuses on the materials starting from lossy compression
  - materials before it are already covered by the previous midterm practices

lossy +

- Recommendation:
  - Bring a calculator will make calculations easier

- Important:
  - Bring a photo ID (Student ID, Driver license, etc.)
  - We need to verify your identity

# Uniform Quantizer

- All bins have the same size except possibly for the two outer intervals: INTERVAL SAME SIZE
  - bi and yi are spaced evenly SPACE EVEN
  - The spacing of bi and yi are both **Δ (step size)**

BOUNDARY BINS ↳ Reconstruction level

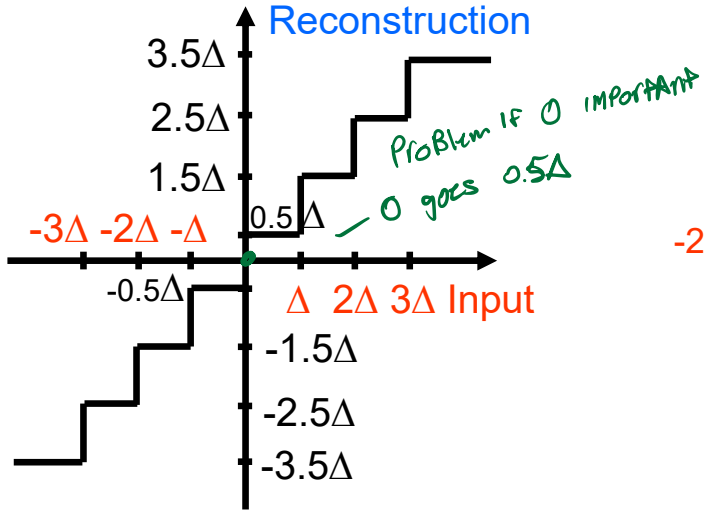
Middle Point

$$y_i = \frac{1}{2}\left(b_{i-1} + b_i\right)$$ for inner intervals. 2 BOUND / 2

ODD BINS CAUSE the 0

Uniform Midrise Quantizer

Even Bins




Problem if 0 important

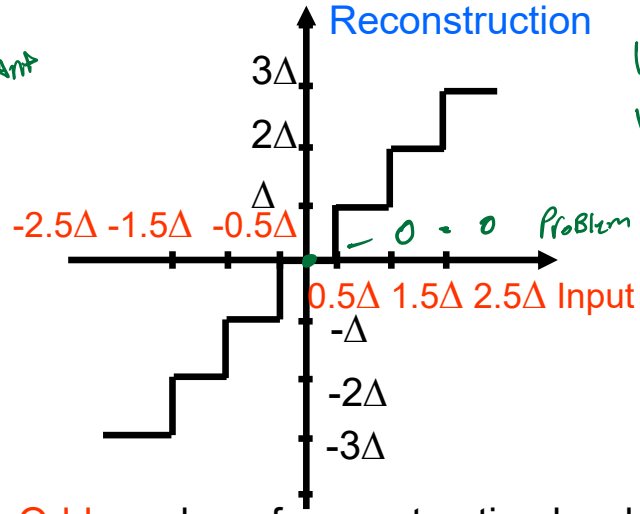
0 goes 0.5Δ

Even number of reconstruction levels
0 is not a reconstruction level

Uniform Midtread Quantizer

Difference

Whether Reconstion Value 0




0 - 0 Problem if 0.1 0.2 important

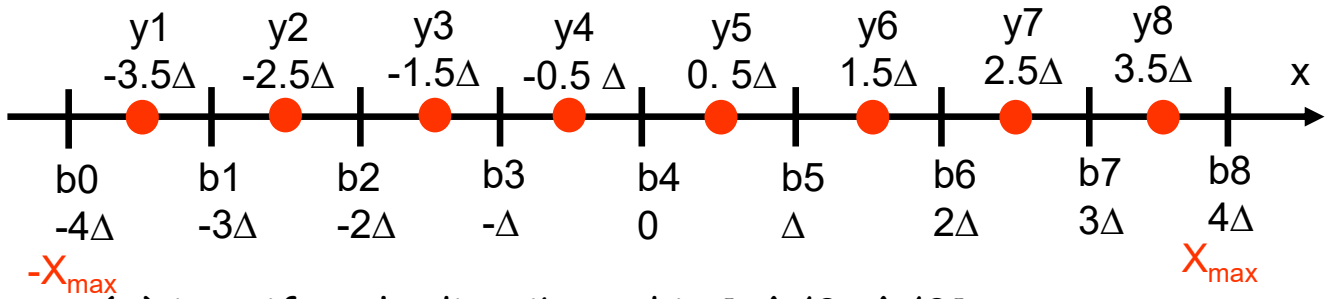
Odd number of reconstruction levels
0 is a reconstruction level

MIDRISE QUANTIZER

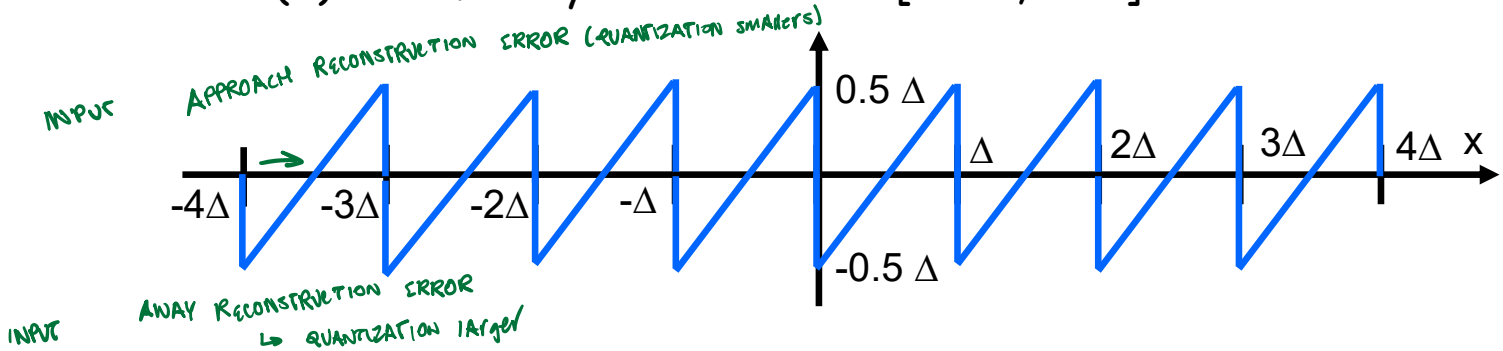
# Uniform Quantization of a Uniformly Distributed Source

SYMMETRIC

- Input X: uniformly distributed in $[-X_{max}, X_{max}]$: $f(x) = 1 / (2X_{max})$ ASSUME INPUT UNIFORM DISTRIBUTE
- Number of bins: M (even for midrise quantizer) TREAT INPUT DISCRETE $X_{MAX} = 1$ $P \frac{1}{2}$
- Step size is easy to get: $\Delta = 2X_{max} / M$. SPACE SAME therefore / BINS
- $b_i = (i - M/2)\,\Delta$




- ➔ e(x) is uniformly distributed in $[-\Delta/2, \Delta/2]$.



## Uniform Quantization of a Uniformly Distributed Source

- MSE

$$MSE_q = \int_{-\infty}^{\infty} (x-\hat{x})^2 f(x)dx = \sum_{i=1}^{M} \int_{b_{i-1}}^{b_i} (x-y_i)^2 f(x)dx$$

$$= M \frac{1}{2X_{\max}} \int_0^{\Delta} \left(x - \frac{\Delta}{2}\right)^2 dx = \frac{M}{2X_{\max}} \frac{1}{12} \Delta^3 = \boxed{\frac{1}{12}\Delta^2}$$

- M increases, Δ decreases, MSE decreases Reduce MSE now By more Bins
- Variance of a random variable uniformly distributed in [- Δ/2, Δ/2]:

$$\sigma^2{}_q = \int_{-\Delta/2}^{\Delta/2} (x-0)^2 \frac{1}{\Delta} dx = \frac{1}{12}\Delta^2$$

- Optimization: Find M such that MSE ≤ D

$$\frac{1}{12}\Delta^2 \le D \Rightarrow \frac{1}{12}\left(\frac{2X_{\max}}{M}\right)^2 \le D \Rightarrow M \ge X_{\max}\sqrt{\frac{1}{3D}}$$

# Signal to Noise Ratio (SNR)

- Variance is a measure of signal energy
- Each bin index is represented by n bits  N = 2 means 4 Bins Assume usage maxed
- So M = $2^n$

Bins

Replace stepsize w interval

$$SNR(dB) = 10\log_{10}\frac{Signal\ Energy}{Noise\ Energy} = 10\log_{10}\frac{1/12(2X_{\max})^2}{1/12\Delta^2}$$ the MSE

$$= 10\log_{10}\frac{(2X_{\max})^2}{(2X_{\max}/M)^2} = \underline{10\log_{10} M^2} = 10\log_{10} 2^{2n} = (20\log_{10} 2)n$$

$$\approx 6.02n\ \ dB$$ cannot utilize if bins not $2^n$ utilize

- If n→n+1, Δ is halved, noise variance reduces to 1/4, and SNR increases by 6 dB.

n + 1 means + SNR 6DB

1 more quantizer bits

# Band Masking: Example

- After analysis, the first levels of 16 of the 32 bands:

| Band | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Level (db) | 0 | 8 | 12 | 10 | 6 | 2 | 10 | 60 | 35 | 20 | 15 | 2 | 3 | 5 | 3 | 1 |

Given

- If the level of the 8th band is 60dB, it gives a masking of 12 dB in the 7th band, 15dB in the 9th.
- Level in 7th band is 10 dB ( < 12 dB ), so ignore it. Anything < 12 ignore
- Level in 9th band is 35 dB ( > 15 dB ), so send it. Anything > 15 encode Difference (35 - 15) = 20

4 Bits cause 6.02(4) = ~20+

[ Only the amount above the masking level needs to be sent, so instead of using 6 bits to encode it, we can use 4 bits -- a saving of 2 bits (= 12 dB). ]

# Discrete Cosine Transform (DCT)

- DCT – close to optimal (known as KL Transform) but much simpler and faster

- Definition:

$$c_{i,j} = a\cos\left(\frac{(2j+1)\,i\,\pi}{2N}\right), \quad \mathrm{i, j} = 0, \ldots, \mathrm{N-1}.$$

$$a = \sqrt{1/N} \quad \text{for } \mathrm{i} = 0,$$

$$a = \sqrt{2/N} \quad \text{for } \mathrm{i} = 1, \ldots, \mathrm{N-1}.$$

- Matlab function:
  - `dct(eye(N));`

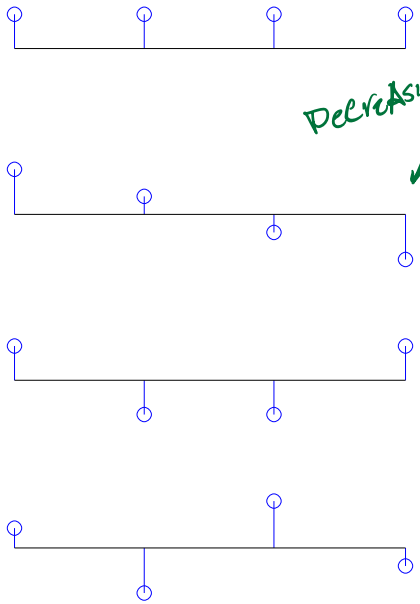
each row transform matrix - 1 frequency

# 4-point DCT

4 x 4 size

constant frequency

decreasing



| | | | | |
|---|---|---|---|---|
| 0.5000 | 0.5000 | 0.5000 | 0.5000 | constant |
| 0.6533 | 0.2706 | -0.2706 | -0.6533 | higher |
| 0.5000 | -0.5000 | -0.5000 | 0.5000 | higher |
| 0.2706 | -0.6533 | 0.6533 | -0.2706 | higher |

$$C = \sqrt{\frac{2}{(4)}} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \cos\frac{\pi}{2(4)} & \cos\frac{3\pi}{2(4)} & \cos\frac{5\pi}{2(4)} & \cos\frac{7\pi}{2(4)} \\ \cos\frac{2\pi}{2(4)} & \cos\frac{6\pi}{2(4)} & \cos\frac{10\pi}{2(4)} & \cos\frac{14\pi}{2(4)} \\ \cos\frac{3\pi}{2(4)} & \cos\frac{9\pi}{2(4)} & \cos\frac{15\pi}{2(4)} & \cos\frac{21\pi}{2(4)} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{\sqrt{2}}\cos\frac{\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{3\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{5\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{7\pi}{8} \\ \frac{1}{\sqrt{2}}\cos\frac{\pi}{4} & \frac{1}{\sqrt{2}}\cos\frac{3\pi}{4} & \frac{1}{\sqrt{2}}\cos\frac{5\pi}{4} & \frac{1}{\sqrt{2}}\cos\frac{7\pi}{4} \\ \frac{1}{\sqrt{2}}\cos\frac{3\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{9\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{15\pi}{8} & \frac{1}{\sqrt{2}}\cos\frac{21\pi}{8} \end{bmatrix}$$

- product DCT & input trying determine input similar which frequencies → similar means ↑ coefficient

# JPEG (Joint Photographic Experts Group)

STUDY

1 Block Preparation

- RGB to YUV (YIQ) planes

2 Transform

- 2D Discrete Cosine Transform (DCT) on 8x8 blocks.

3 Quantization

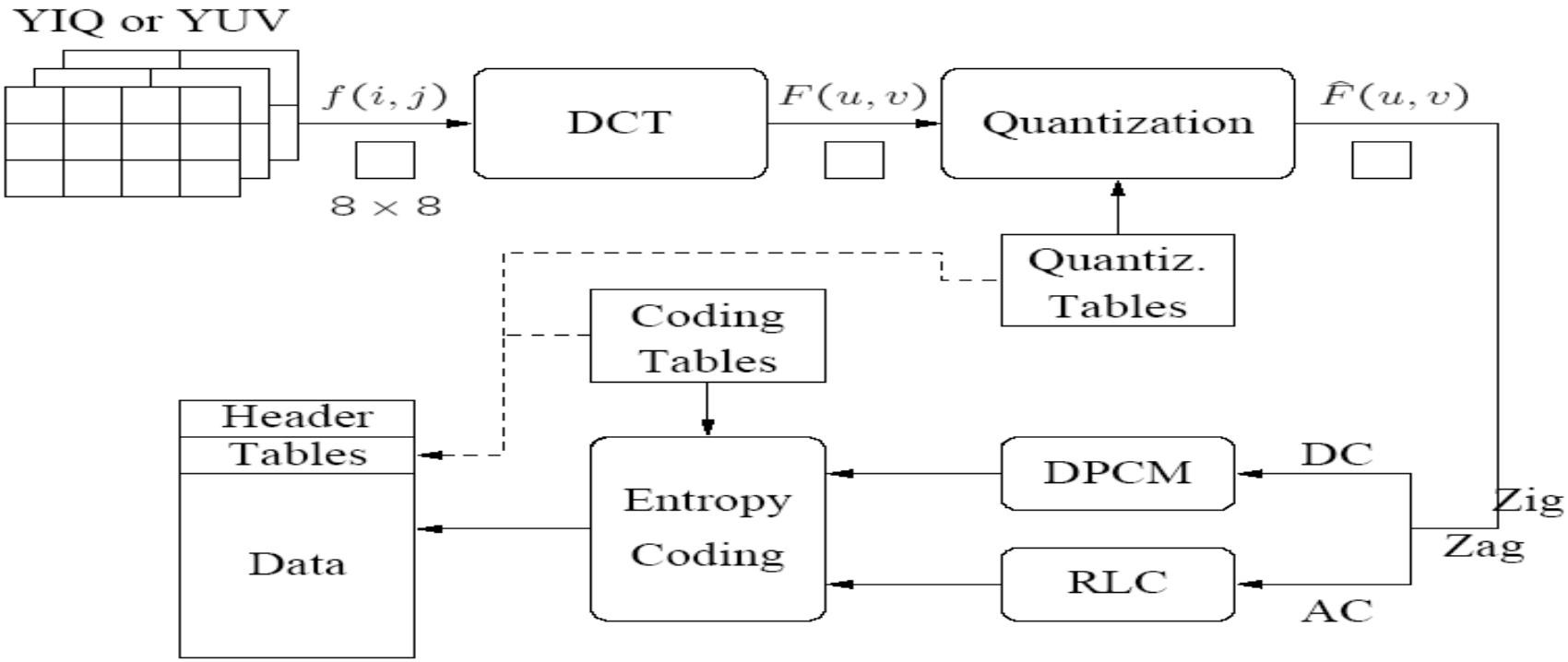
- Quantized DCT Coefficients (lossy).

4 Encoding of Quantized Coefficients

- Zigzag Scan
- Differential Pulse Code Modulation (DPCM) on DC component
- Run Length Encoding (RLE) on AC Components
- Entropy Coding: Huffman or Arithmetic

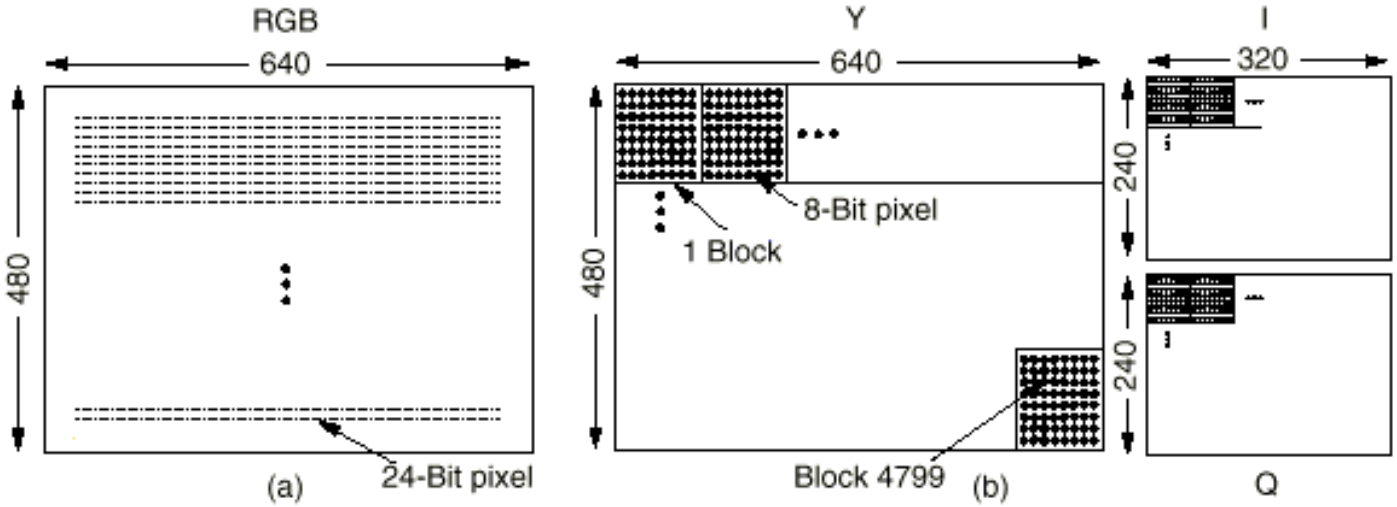
# JPEG Diagram

STUDY



# JPEG: Block Preparation

Doesn't compress everything @ once
↳ divide into block compress those




Chroma subsampling JPEG means I & Q sizes smaller.

**RGB Input Data** **After Block Preparation**

**Input image: 640 x 480 RGB (24 bits/pixel) transformed to three planes:**

**Y: (640 x 480, 8-bit/pixel) Luminance (brightness) plane.**
**U, V: (320 X 240 8-bits/pixel) Chrominance (color) planes.**

# Block Effect

JPEG compresses Blocks independently

- Using blocks, however, has the effect of isolating each block from its neighboring context.
  - choppy (“blocky") with high *compression ratio*

**Compression Ratio: 7.7**

**Compression Ratio: 33.9**

**Compression Ratio: 60.1**

# JPEG: Quantized DCT Coefficients

**Uniform quantization:**
**Divide by constant N and round result.**

**In JPEG, each DCT F[u,v] is divided by a constant q(u,v).**

**The table of q(u,v) is called quantization table.**

Quantization table

**q(u,v)**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 2 | 4 | 8 | 16 | 32 | 64 |
| 1 | 1 | 2 | 4 | 8 | 16 | 32 | 64 |
| 2 | 2 | 2 | 4 | 8 | 16 | 32 | 64 |
| 4 | 4 | 4 | 4 | 8 | 16 | 32 | 64 |
| 8 | 8 | 8 | 8 | 8 | 16 | 32 | 64 |
| 16 | 16 | 16 | 16 | 16 | 16 | 32 | 64 |
| 32 | 32 | 32 | 32 | 32 | 32 | 32 | 64 |
| 64 | 64 | 64 | 64 | 64 | 64 | 64 | 64 |

DCT Coefficients

**F[u,v]**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 150 | 80 | 40 | 14 | 4 | 2 | 1 | 0 |
| 92 | 75 | 36 | 10 | 6 | 1 | 0 | 0 |
| 52 | 38 | 26 | 8 | 7 | 4 | 0 | 0 |
| 12 | 8 | 6 | 4 | 2 | 1 | 0 | 0 |
| 4 | 3 | 2 | 0 | 0 | 0 | 0 | 0 |
| 2 | 2 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Quantized coefficients

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 150 | 80 | 20 | 4 | 1 | 0 | 0 | 0 |
| 92 | 75 | 18 | 3 | 1 | 0 | 0 | 0 |
| 26 | 19 | 13 | 2 | 1 | 0 | 0 | 0 |
| 3 | 2 | 2 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Rounded F[u,v]/ Q(u,v)**

Coefficients divide by quantization table round result → different than lossy
Reconstruct → table

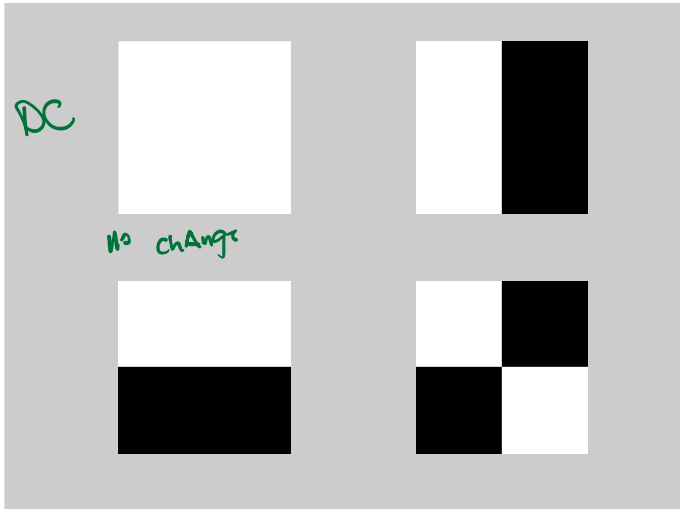
# More about Quantization

- **Quantization is the main source for loss**
  - *Q*(*u, v*) of larger values towards lower right corner
    - More loss at the higher spatial frequencies
    - Supported by Observations 1 and 2.
  - Q(u,v) obtained from psychophysical studies
    - maximizing the compression ratio while minimizing perceptual losses

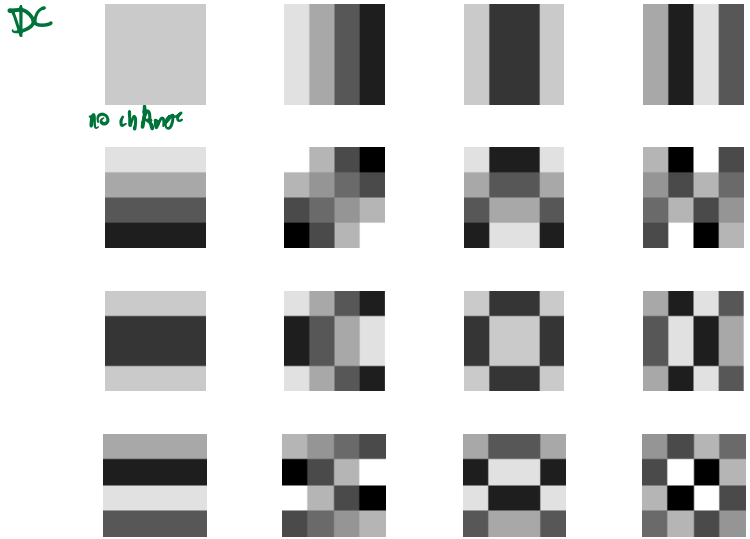
cause BR higher frequency
↳ least important eyes cannot determines
↳ Below 64 rounds 0
↳ ignore

# Recall: 2-D DCT Basis Matrices

For 2-point DCT




For 4-point DCT

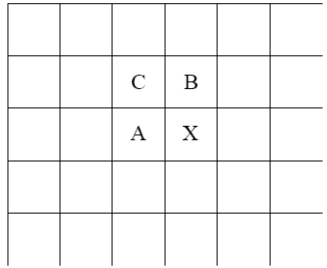


# Lossless Version of JPEG

- **A special mode; different from the standard JPEG**
  - Using prediction and entropy coding only
- **Forming a differential prediction:**
  - A predictor combines the values of up to three neighboring pixels as the predicted value for the current pixel
  - 7 schemes for combination
- **Encoding:**
  - The encoder compares the prediction with the actual pixel value at the position `X' and encodes the difference using entropy coding

# 7 Predictors

| Predictor | Prediction |
|---|---|
| P1 | A |
| P2 | B |
| P3 | C |
| P4 | A + B - C |
| P5 | A + (B - C) / 2 |
| P6 | B + (A - C) / 2 |
| P7 | (A + B) / 2 |




P1 6
P2 -3
P3 -1
P4 4
P5 5
P6 0.5
P7 1.5

closer 0 greater

110 112
103 109

i.e., C=110, B=112, A=103, X=109

# Final Review Strategy

- Study the slides
- Memorizing things usually does not work
  - Question wordings may change in exams
- Study the assignment questions
- Understanding materials and pay attention to details

Per Symbol And B Bytes B Bits